

Method Report — Billing Event Reconstruction and Payment-Delay Early-Warning for Clinical Laboratory Settings

– Version: 2.0

– Date: 2026-09-22

1. Research Questions

– RQ1: Under multi-source, missing or conflicting records, does uncertainty-preserving event reconstruction (explicit lineages, conflicts, needs-review queue) reduce wrong links and control manual review volume, compared with simple identifier joins and appropriate existing methods?

– RQ2: Does incorporating available record-quality and processing-stage signals improve within-horizon prediction of first-observable positive payment allocation vs. no-payment closure vs. still-open, relative to aging-rule and basic baselines?

2. Related Work and Position

– Fellegi & Sunter (1969): decision-theoretic record linkage with a possible-link disposition; the needs-review tier operationalizes this disposition.

– Christen (2012) and van der Aalst (2016): general matching and event-log analysis; this work adds billing-domain event semantics and as-of point-in-time discipline.

– Deep Claim (arXiv:2007.06229): predicts payer responses pre-submission from claims; this work differs in scope: post-submission billing-event reconstruction plus an observation-window-aware competing-event forecast.

– Fine & Gray (1999) / Kaplan & Meier (1958): competing-risk and censoring foundations; a discrete-time interval formulation is adopted for irregular observation.

– Zeng et al. (2007), Schoonbee (2022): invoice-lateness prediction; the aging baseline is informed by these, with explicit censoring treatment added.

Full scan: research/prior_work.csv (nine references, verified).

3. Data and Permissions

- The study uses de-identified billing event records from three participating clinical laboratory organizations, referred to by site tokens CPL, TRI and NDX under the data-use arrangement. Site tokens are used throughout; organizational identities are recorded in the study documentation.
- Data window: 2022-07-01 to 2023-11-15. Records carry per-site local timezones (including daylight-saving transitions); submission and payment activity follows weekday-dominant rhythms with a mix of same-day and lagged record entry.
- Data quality issues observed and handled: duplicate imports, cross-organization duplicate bill numbers, missing correction references, timezone-less timestamps, invalid amounts and non-USD currencies. These are isolated by the schema validator, never silently repaired, and reported in the run documentation.
- Manually adjudicated reference labels (record pair lineage membership and observed outcome times) are held out from all model inputs and used only for evaluation.
- Real-mode processing of identified data remains disabled; see `governance/data_permissions.json`.

4. Event Dictionary and Prediction Target

- Event types: SUBMITTED, CORRECTED, PAYMENT_POSTED, PAYMENT_REVERSED, VOID_VERSION, CLOSED_NO_PAYMENT, REOPENED, ADJUSTMENT, STATUS_UPDATED, UNKNOWN (with review).
- Target: within $H=30$ days after landmark t (first observable submission + 7 days), predict the first observable positive payment allocation, first no-payment closure, or still-open. Not final bank cash, not full settlement.
- Protocol v1.1 (PC01, 2024-08-06): interval granularity 1 -> 5 days after sparsity findings at 1-day resolution.

5. Candidate Methods

- Module A: A0 exact-id baseline (native `previous_record_id` with org/cycle/time checks);

A1 constrained linkage (R2 same-bill-id version chains; R3 amount/date-window weak evidence with uniqueness gating). Accepted / rejected / needs-review tiers; no unconstrained transitive chaining; manual decisions applied as a separate layer.

- Module B: B0 age/stage empirical-risk baseline (abstains on unsupported cells);

B1 discrete-time multinomial logistic regression; C1 B1 plus reconstruction-quality features. Class probabilities per interval with the product-rule cumulative curves; sum-to-one enforced (T16).

6. Splits and Leakage Control

- Temporal split grouped by lineage, 60/20/20; test portion locked.

- Features computed strictly from records observable at t (T10); reference labels and outcome fields excluded from features (T13); lineages never span splits (T14); preprocessing fitted on train only (T15).

7. Results (locked test set)

Module A (held-out adjudicated adjacency evaluation):

- reference pairs 95; predicted 123
- true positives 94; false positives 29;

false negatives 1

- precision 0.7642; recall 0.9895; refusal ratio 0.0956
- The weak-evidence rule R3 is the dominant false-positive source; multi-candidate

ambiguity is routed to review instead of being forced.

Module B (snapshot-level, n=206 test snapshots; outcome mix

{"payment": 539, "still_open": 488, "censored": 2, "close_no_payment": 8}):

- B0: snap log-loss 0.736, Brier 0.514
- B1: snap log-loss 0.669, Brier 0.462

(relative improvement 9.0%)

- C1: snap log-loss 0.668, Brier 0.462

(delta vs B1 -0.0009 — no incremental value)

8. Negative / Neutral Findings (reported as-is)

- C1 quality features add no measurable value on the locked test set.
- An early implementation with balanced class weighting distorted conditional

probabilities approximately 4x; corrected after diagnostics (2024-12).

- Close-no-payment events are rare in the observation window (n=8);

the model abstains rather than fabricating probabilities where support is insufficient.

9. Applicability and Limitations

- Applicable condition: multi-source billing exports with availability-time semantics,

USD-denominated amounts, explicit or inferable version chains.

- Not applicable: final collection forecasting, cross-currency aggregation,

identified patient-level analytics without authorization.

- Limitations: the participating sites span three ownership models in one national

context; generalization to other settings requires further validation. Landmark

and horizon are fixed study defaults pending operational confirmation. No external

independent validation has been completed to date.

10. Contributions

- The research design, decision records (research/decisions/, D01-D06 and PC01) and

the analysis were carried out by the applicant; engineering execution used

computational tooling under her direction. Individual decisions trace to code and

tests as documented in the decision records.

11. Reproducibility

- Environment: environment.json (Python 3.12.13; historical snapshot 3.11.5).

- Commands: the standard pipeline entry (full pipeline); pytest (22 acceptance

tests); the validation entry (held-out evaluation). Full command references in README.

- Code and data access: <https://github.com/fanshiyu35/lab-billing-research>

- Run referenced: the v1.4 reference run (artifacts archived with the code release).

References

See research/prior_work.csv for the verified reference list.